Angles and the Unit Circle

Name _____

Algebra 2 Lesson 7-1 (fast track) Date _____ Period _____

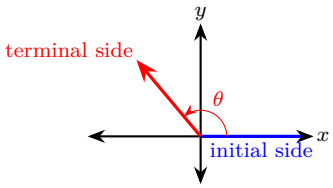
Learning goal: I can find a positive, a negative, and a greater-than-360° measure for an angle in standard position, and convert between degrees and radians.

Language goal: I can explain where an angle ends using the words *initial side*, *terminal side*, *quadrant*, *coterminal*, and *radian*.

On the quiz: (1) You will pick every measure, in degrees or radians, that names one drawn angle (Examples 1 and 3). (2) You will pick the positive and negative measures of an angle from its reference angle and quadrant (Example 1; full Savvas quiz only). (3) You will find the angle in standard position from how far its terminal side is from the x-axis and its quadrant (Example 1). (4) You will convert radians to degrees (Example 3). (5) Arc length on a circle (Example 4, not in this packet; full Savvas quiz only).

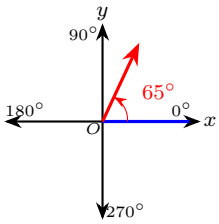
Example 1: Explore Angles in the Coordinate Plane

A. How can you express angle measures in the coordinate plane? An angle is in **standard position** when its vertex is at the origin and its **initial side** is the positive x-axis. The **terminal side** is the other ray that forms the angle. An angle θ is measured **counterclockwise** from its initial side to its terminal side. The terminal side tells you the **quadrant**:

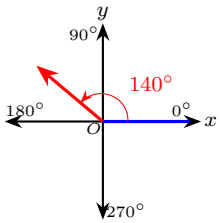


$0° < \theta < 90°$	Quadrant I	$180° < \theta < 270°$	Quadrant III
$90° < \theta < 180°$	Quadrant II	$270° < \theta < 360°$	Quadrant IV

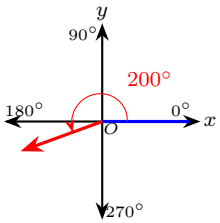
How far is each terminal side from the x-axis? The quiz calls the smallest angle between the terminal side and the x-axis the **reference angle**.



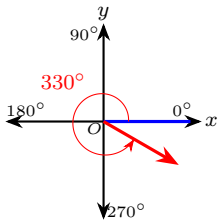
65°: Quadrant I
65° above the $+x$-axis



140°: Quadrant II
$180° - 40°$



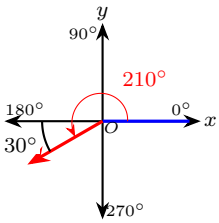
200°: Quadrant III
$180° + 20°$

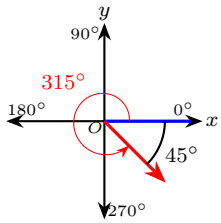


330°: Quadrant IV
$360° - 30°$

B. Sketch a 210° angle and a 315° angle.

- $210° = 180° + 30°$. The terminal side is 30° beyond the negative x-axis, in Quadrant III.
- $315° = 360° - 45°$. It is a full revolution minus 45°, in Quadrant IV.





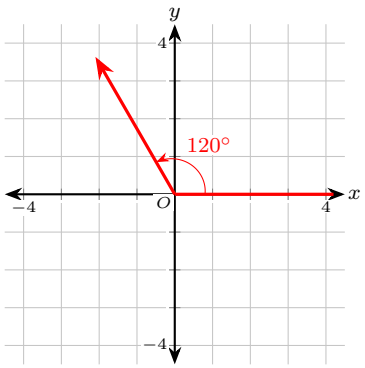
C. How can you represent negative angle measures and angles with measures greater than 360°?

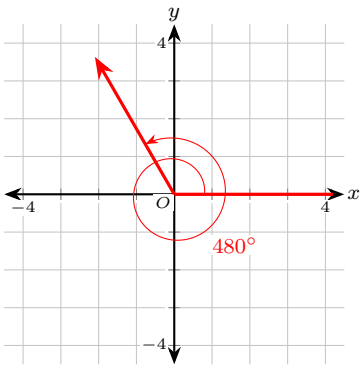
Positive: $0° \le \theta < 360°$

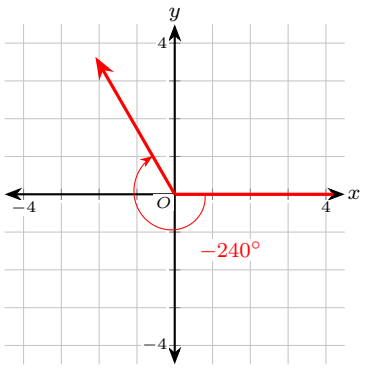
Greater than 360°**:** $\theta + 360°k$, where k is the number of full rotations

Negative: $-360° \le \theta < 0°$, measured **clockwise**

$120° + 360° = 480°$ (one more turn)
$120° - 360° = -240°$ (clockwise)
All three share the same terminal side, so they are **coterminal angles**.





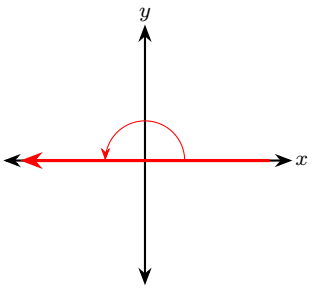


Say it: The angle 210° has its terminal side in Quadrant _____, _____° beyond the negative x-axis. Two angles that are coterminal with it are _____° and _____°, because . . .

Try It 1: Positive, negative, and greater than 360°

Find a positive angle measure, a negative angle measure, and an angle measure greater than 360° for each angle.

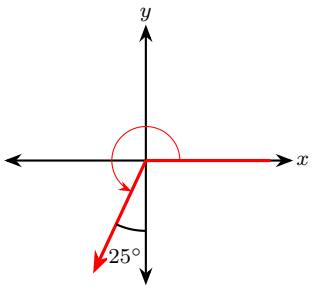
a.



Positive: _____ Negative: _____

Greater than 360°: _____

b.



Positive: _____ Negative: _____

Greater than 360°: _____

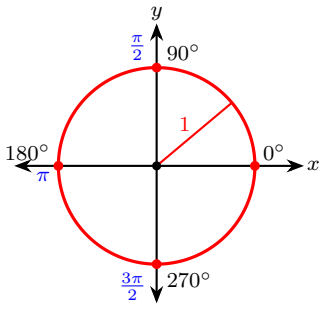
Hint: Trace each angle with your finger first. A positive measure turns counterclockwise; a negative measure turns clockwise.

Example 3: Convert Between Degrees and Radians

One full rotation, or 360°, is the same as 2π **radians**. Half a rotation, or 180°, is the same as π radians.

Radians	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
Degrees	0	90	180	270	360

Radians are written without a unit; degrees always get the ° sign.



A. Convert $\dfrac{5\pi}{4}$ into degrees.

1. 2π radians $= 360°$, so π radians $= 180°$.
2. Divide both sides by π: 1 radian $= \frac{180°}{\pi}$.
3. Multiply by $\frac{180°}{\pi}$:

$$\frac{5\pi}{4} \cdot \frac{180°}{\pi} = \frac{5 \cdot 180°}{4} = 225°$$

Answer: $\frac{5\pi}{4}$ radians $= 225°$.

Check: 225° is between 180° and 270° (Quadrant III), just as $\frac{5\pi}{4}$ is between π and $\frac{3\pi}{2}$.

B. Convert 75° into radians.

1. π radians $= 180°$.
2. Divide both sides by 180: $1° = \frac{\pi}{180}$ radians.
3. Multiply by $\frac{\pi}{180°}$:

$$75° \cdot \frac{\pi}{180°} = \frac{75\pi}{180} = \frac{5\pi}{12}$$

Answer: $75° = \frac{5\pi}{12}$ radians.

Simplify $\frac{75}{180}$ by dividing both by 15.

Say it: To change radians to degrees, I multiply by _____ because π radians = _____°. The angle $\frac{5\pi}{4}$ radians is _____°, so its terminal side is in Quadrant _____.

Try It 3: Convert

a. Convert $\dfrac{\pi}{4}$ and 1.5 to degrees. Round to the nearest tenth.

$\frac{\pi}{4}$ = _____° 1.5 = _____°

b. Convert 30° and 115° to radians.

30° = _____ 115° = _____

Hint: 1.5 has no π in it, but it is still a radian measure. Multiply it by $\frac{180°}{\pi}$ the same way.

IXL: Practice for the quiz

Go to IXL, Algebra 2. Type the code in the search box. Do the starred skills first.

Code	Skill	Feeds
ANN	Quadrants	Warm-up before Example 1
7CV ★	Coterminal angles	Quiz 1 (bonus): positive, negative and greater-than-360° names (Example 1); Quiz 2 (full Savvas quiz only)
BRP ★	Reference angles	Quiz 3: the angle from its reference angle and quadrant (Example 1); Quiz 2 (full Savvas quiz only)
EDC ★	Convert between radians and degrees	Quiz 4, and the radian choices in Quiz 1 (Example 3)
UA5	Radians and arc length	Optional: Quiz 5, arc length (Example 4, not in this packet; full Savvas quiz only)

SmartScore 70 or above on each starred (★) skill replaces your quiz score.

Exit question: A positive and a negative angle

Generalize What is the relationship between a positive angle and a negative angle that share a common terminal side? Write a formula that relates the two measures.

Hint: Try it with Example 1C first: 120° and −240°.

Before the quiz, finish these sentences.

An angle is in standard position when its vertex is at the _____ and its initial side is on the _____.

Coterminal angles share the same _____ side. I find one by adding or subtracting _____°.

A terminal side in Quadrant II that is 40° from the negative x-axis makes the angle _____°.

A terminal side in Quadrant IV that is 30° from the positive x-axis makes the angle _____°.

To change degrees to radians, I multiply by _____. One full rotation is _____ radians.